

Experimental identification of river-temperature information regimes

Core design: 2 forcing levels x 2 local-information levels x 2 geometries x 2 model classes x 3 leads = 48 matched cells

(a) Four experimental axes

F - Future forcing

F0 no future meteorology
F3 realized future meteorology
retrospective oracle

L - Local information

L0 history-rich gauged site
L2 no target-site WT history
L1 / L2-U2: diagnostics

G - Spatial geometry

Random-site
Whole-region

A - Model class

LightGBM
Plain causal TCN
ThermoRoute: selected extension

Forecast leads

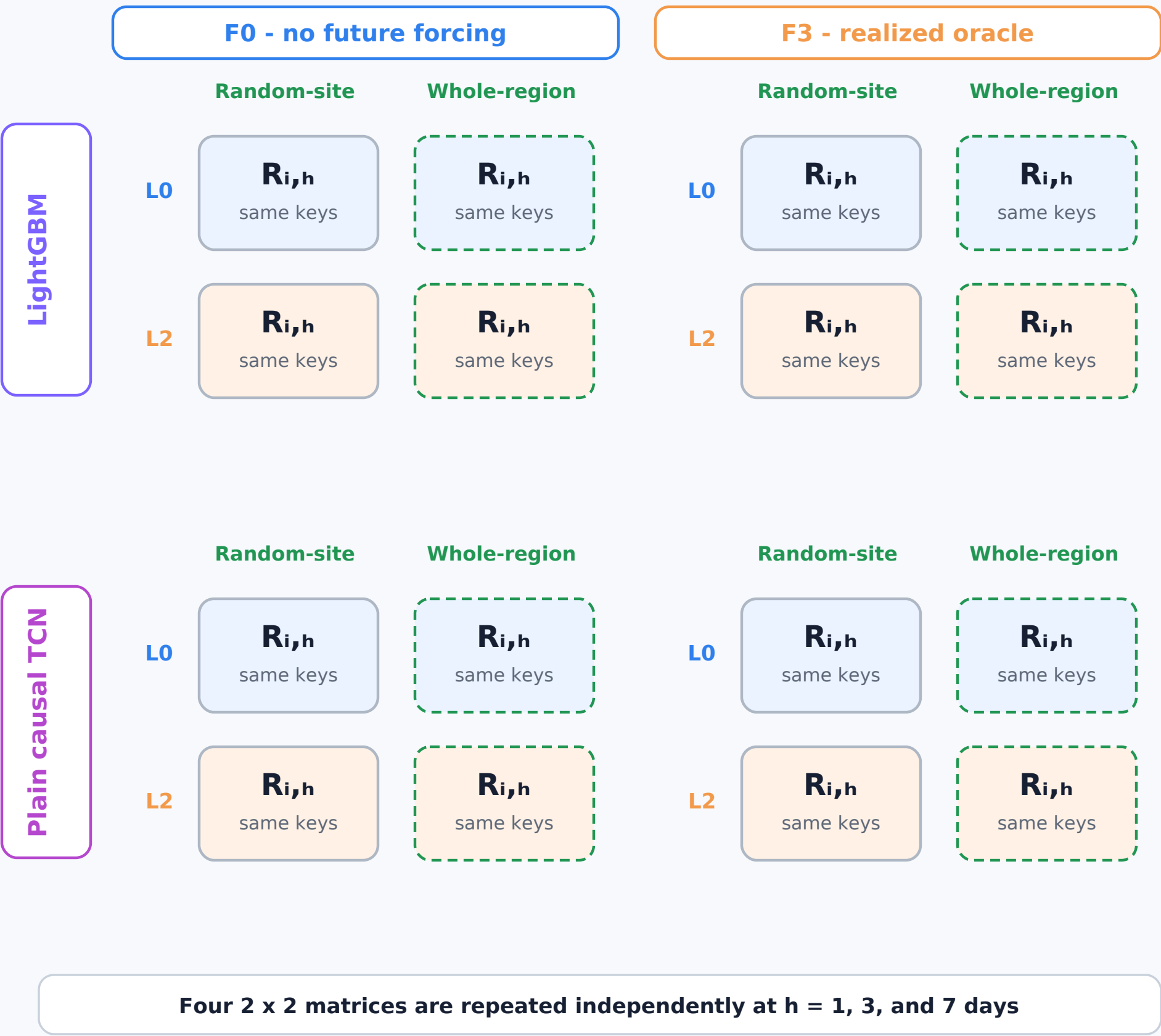
1 day

3 days

7 days

Every factor is defined by an information contract, not by a model-specific feature set.

(b) Core matched response surface



(c) Identified contrasts

Future-forcing value

$$V_F = R(F0) - R(F3)$$

Local-state value

$$V_L = R(L2) - R(L0)$$

Geometry penalty

$$P_G = R(\text{region}) - R(\text{random})$$

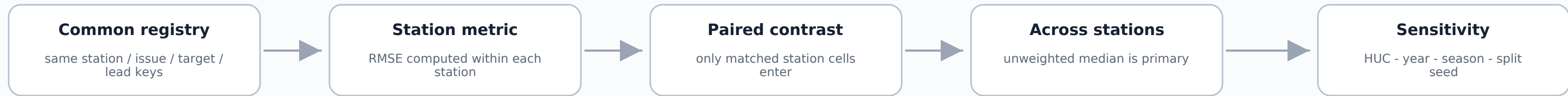
Architecture value

$$V_A = R(\text{LGBM}) - R(\text{TCN})$$

Double differences

- F x L** Does forcing value change without local WT?
- F x A** Does a sequence model use future forcing better?
- L x G** Does local WT mask spatial-transfer difficulty?
- L x A** Does architecture matter in cold-start transfer?

(d) One scoring contract for every cell and contrast



F3 is a retrospective realized-meteorology oracle; it is not an operational forecast.